

Problems marked with a (*) are a bit harder. Hints are available in the final page.

1.1. Recap Questions.

1. Let $\{x_k\}_{k \in \mathbb{N}} \subset [1, 2] \subset \mathbb{R}$ be a sequence. Is it necessarily convergent?
2. Describe all differentiable functions $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ such that $f'(x) = 1/x$ for all $x \neq 0$.
3. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous and bounded, is it true that

$$\int_0^{\pi/2} f(\cos(x)) dx = \int_0^1 \frac{f(t)}{\sqrt{1-t^2}} dt?$$

4. Let $f: [0, \infty) \rightarrow \mathbb{R}$ have $f''(x) > 0$ for all $x \geq 0$. Can f have a global maximum point?
5. Let $\phi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear map and let $Q := [0, 1] \times [0, 1]$ be the unit square. How does $\phi(Q)$ look like? What are the possibilities?
6. Consider the vector space V of polynomials of degree less than 4, and the function $D: V \rightarrow V$ which associates to a polynomial p its derivative p' . Is D a linear map? If so, what is the dimension of its null space?

1.2. From equations to drawings. Sketch the following subsets of $\mathbb{R}^2, \mathbb{R}^3$.

1. $A := \{(x, y) : x^2 + y^2 < 1, y \geq x^2\}$
2. $B := \{(x, y) : y^2 = x^3 - x\}$
3. $C := \{(x, y) : \frac{\pi}{6} < \arctan(y/x) < \frac{\pi}{3}, 0 < x < y\}$
4. $D := \{(x, y, z) : \sqrt{x^2 + y^2} = 1 - z, 0 < z < 1\}$
5. $E := \{(x, y, z) : z = \sqrt{1 + x^2 + y^2}\}$
6. $F := \{(x, y, z) : z = \max\{|x|, |y|\}\}$

1.3. From drawings to equations. Express in terms of functions and equations the following sets.

1. The coordinates of a point in the plane that goes endlessly back and forth from the origin to the point $(-1, 1)$.
2. A planar domain with two holes in it.
3. A bi-dimensional disk sitting inside the plane $\{x + y + z = 0\}$ in $\mathbb{R}^3$.
4. A Pac-Man shaped domain in the plane.
5. A pyramid, whose base is a square, sitting in $\mathbb{R}^3$.

1.4. Young's inequality. Let $a, b > 0$, and let $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$. The goal of this exercise is to prove that $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$.

Obs: The case $p = q = 2$ corresponds to the well-known inequality $ab \leq \frac{a}{2} + \frac{b}{2}$, which is just a rewriting of $(a - b)^2 \geq 0$.

1. Prove that for all $t > 0$,

$$t^{1/p} \leq \frac{t}{p} + \frac{1}{q}.$$

2. Multiply the inequality by b^q and choose an appropriate value of t to conclude.

A bonus: Show that $\langle a, b \rangle \leq \frac{\|a\|^p}{p} + \frac{\|b\|^q}{q}$ for all $a, b \in \mathbb{R}^n$.

Hints:

- 1.2 As ETH students you have a license for Mathematica. All these plots are one-line-commands using that software!
- 1.2.2 The set is clearly symmetric with respect to $y \mapsto -y$, so it's enough to plot qualitatively the function $y(x) = \sqrt{x^3 - x}$, separately in the intervals $x \in [-1, 0]$ and $x \in [0, \infty)$... From here you can use all the technology from Analysis I...
- 1.2.4 Try to figure out how D looks layerwise, i.e., as a planar domain once z is kept fixed.
- 1.2.5 If you know how to plot $z = f(s), s \geq 0$, then the set $\{(x, y, z) : z = f(\sqrt{x^2 + y^2})\}$ is obtained rotating the plot of f around the z -axis...
- 1.3.1 Construct an appropriate function $f: \mathbb{R} \rightarrow \mathbb{R}^2$, where t represents the time and $f(t)$ represents the position of the moving point at time t . You want f to be periodic in t and $f(t)$ to be aligned with the direction $(-1, 0)$
- 1.3.3 Observation: the intersection of a round ball with a plane is a disk...
- 1.3.5 Take inspiration from 1.2.4 and 1.2.6...
- 1.4.1 Defining $g(t) = \frac{t}{p} + \frac{1}{q} - t^{1/p}$, this is equivalent to $g(t) \geq 0$. Compute $g'(t)$.